Seat No.: _____ Enrolment No._____

GUJARAT TECHNOLOGICAL UNIVERSITY

MCA INTEGRATED– SEMESTER-II EXAMINATION –SUMMER-2020

Subject Code:2628601 **Date:04-11-2020**

Subject Name: Data Structures (DS)

Time:10:30 AM to 01:00 PM **Total Marks: 70**

Instructions:

1. **Attempt all questions.**
2. **Make Suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Q.1	**(a)**	1. Differentiate complete binary tree and full binary tree 2. Define: Directed Graph 3. In binary tree maximum no. of nodes on level 1 is _____. 4. Define: Algorithm 5. List out applications of stack. 6. Define: Spanning Tree. 7. Define: Leaf Node	**07**
	(b)	Define data structure. Explain different types of data structures.	**07**
Q.2	**(a)**	What is Queue? Write an algorithm for inserting and deleting an element into circular queue.	**07**
	(b)	Explain Tower of Hanoi problem by moving 3 discs.	**07**
		OR	
	(b)	Write an algorithm for conversion of given infix expression into postfix expression using stack.	**07**
Q.3	**(a)**	What is Linked list? Explain types of linked list in detail.	**07**
	(b)	Write an algorithm for binary search.	**07**
		OR	
Q.3	**(a)**	Write an algorithm to insert and delete node from singly linked list.	**07**
	(b)	Define searching. Write an algorithm for linear search.	**07**
Q.4	**(a)**	Find preorder: Inorder : 1 3 5 6 4 2 7 Postorder : 6 5 4 3 7 2 1	**07**
	(b)	What is sparse matrix? Explain multi-linked structure for sparse matrix with example.	**07**
		OR	
Q.4	**(a)**	Draw expression tree of given expression : A * B + C – D / E * F ^ G	**07**
	(b)	Explain threaded binary tree with example.	**07**
Q.5	**(a)**	Explain Quick sort with example.	**07**
	(b)	Write DFS algorithm and explain it with example.	**07**
		OR	
Q.5	**(a)**	What is hashing? Explain various techniques for collision resolution with appropriate example.	**07**
	(b)	Explain 2-3 tree with example.	**07**

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